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CHAPTER 1

INTRODUCTION

1.1 Background

Mental health is an essential component of overall well-being, influencing how individuals think, feel, and act. Despite growing awareness, mental health challenges such as anxiety, depression, and stress continue to affect millions worldwide. The stigma surrounding mental health, coupled with limited access to timely and affordable care, often prevents individuals from seeking the support they need. Traditional therapeutic approaches, while effective, depend on the availability of trained professionals and in-person consultations, which may not always be feasible.

With the advent of Artificial Intelligence (AI) and Natural Language Processing (NLP), there is a growing opportunity to provide scalable and immediate support through digital solutions. Conversational agents, or chatbots, offer an accessible and stigma-free platform for individuals to express their concerns and receive empathetic guidance. These systems can serve as the first point of contact, providing emotional support, promoting self-awareness, and encouraging further professional consultation when needed.

1.2 Problem Statement

Access to mental health services remains a significant challenge, particularly in under-resourced areas or among individuals reluctant to seek help due to societal stigma. The lack of immediate, non-judgmental, and private support often results in emotional distress being ignored or misunderstood. This delay in addressing mental health issues can exacerbate conditions and lead to long-term consequences on personal and professional life.

Furthermore, existing support systems are not always available 24/7, and the cost or availability of professional help can be prohibitive. There is an urgent need for a digital, low-cost, and privacy-conscious alternative that can provide basic mental health support, emotional relief, and early detection of psychological distress. Such a solution would act as a complementary tool alongside traditional mental health services.

1.3 Objectives

The *Thrive* Mental Health Wellbeing Chatbot project aims to tackle these challenges by offering an AI-powered conversational platform that delivers mental health support in real-time. The key objectives of this project include:

- **Building a Responsive Chatbot:** Develop an intelligent chatbot using deep learning and NLP techniques to engage users in empathetic and supportive conversations related to their mental well-being.
- **Emotion and Context Recognition:** Implement models capable of understanding user sentiment, detecting emotional distress, and providing contextually appropriate responses. This includes basic keyword-based triaging as well as generative responses for open-ended input.
- **Model Evaluation:** Assess the chatbot's effectiveness using diverse user scenarios and evaluation metrics such as response relevance, emotional tone matching, and user satisfaction, ensuring that the system maintains empathy and coherence.
- **Practical Deployment:** Deploy the chatbot through a user-friendly web interface using frameworks like Gradio . The interface will allow users to initiate conversations, access mental health resources, and receive motivational or informational feedback.
- **Scalability and Safety:** Ensure the platform supports future extensions such as multilingual capabilities, crisis response routing, integration with professional networks, and compliance with ethical AI guidelines.

1.4 Scope

The *Thrive* Mental Health Wellbeing Chatbot focuses on providing conversational support for mental well-being through text-based interaction. The current scope includes managing mild to moderate emotional issues such as stress, anxiety, loneliness, and motivational concerns. It does not diagnose mental health conditions or replace professional therapy, but it can act as a supportive tool and early touchpoint for users in need of guidance.

The system leverages NLP and pre-trained transformer models for understanding and generating context-aware responses. High-risk or crisis-related queries are redirected

with predefined alerts and suggested helpline contact options. The current implementation is designed for English-language support, with future expansions planned for multilingual support, integration with healthcare apps, and voice-based interactions.

This project lays the groundwork for AI-assisted mental health tools, aiming to bridge the gap between emotional need and immediate support. Through accessibility, empathy, and intelligent response generation, *Thrive* Mental Health Wellbeing Chatbot seeks to promote mental well-being and reduce barriers to mental health care for individuals across diverse communities.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The development of mental health chatbots has progressed significantly, evolving from rule-based systems to AI-driven conversational agents capable of understanding and responding to user emotions in real-time. This section reviews key advancements, existing limitations, and emerging research directions in the field of AI-powered mental health support systems.

2.2. Advancements in AI-Based Chatbots

From Rule-Based Systems to Deep Learning Models

Initial chatbot systems in mental health, such as ELIZA, relied on simple pattern matching and rule-based scripts to simulate empathy. While they demonstrated early potential, their responses were limited and lacked contextual understanding. With the introduction of Natural Language Processing (NLP) and Deep Learning—especially transformer-based models like BERT and GPT—a new generation of chatbots emerged that can understand sentiment, context, and even detect emotional distress in open-ended conversations ([PMC10242473](#), [PMC10360018](#)).[1][2]

Emotion Recognition and Sentiment Analysis

Modern mental health chatbots incorporate sentiment analysis and emotion detection to personalize conversations. These systems classify user input into emotional categories (e.g., sadness, anxiety, joy) to provide empathetic and relevant replies. Research suggests that combining textual and contextual cues improves emotional alignment, which in turn enhances user engagement and trust ([IEEE 10652390](#)).

Conversational Flow and Context Maintenance

Advanced chatbots use sequence-to-sequence models and attention mechanisms to maintain coherent and meaningful dialogues over multiple turns. These capabilities allow chatbots to follow up on earlier parts of a conversation and provide relevant suggestions or coping strategies, reducing the chance of irrelevant or repetitive responses ([IEEE 10170573](#), [IEEE 10489148](#)).

Data Preprocessing and Language Normalization

Mental health datasets often contain informal language, misspellings, and emotionally loaded expressions. NLP preprocessing steps—such as lemmatization, sentiment tagging, and context-aware embeddings—play a critical role in improving model accuracy. Datasets like CLPsych and DAIC-WoZ are commonly used benchmarks for training and evaluation ([IEEE 9544838](#)).

Ensemble and Hybrid Models

Recent studies explore combining rule-based and neural approaches to balance reliability with contextual understanding. Hybrid models are particularly useful in sensitive domains like mental health, where response appropriateness is as important as linguistic correctness ([IEEE 10150665](#), [IEEE 7962482](#)).

2.3 Challenges and Research Directions

- **Ethical Considerations and Privacy:** Ensuring user data protection and building trust are fundamental. Future systems must integrate data encryption, informed consent, and clear boundaries for data usage.
- **Limited Annotated Data:** A major bottleneck in mental health chatbot training is the scarcity of labeled, real-world conversational data. Most datasets are small or synthetic, limiting generalization.
- **Bias and Fairness:** Large language models can inherit societal biases. Researchers are exploring fairness-aware training methods and model auditing to reduce harm and ensure inclusivity ([IEEE 10560327](#)).

- **Interpretability:** For mental health applications, it is important that chatbot responses are explainable. Users and professionals need to understand the logic behind certain responses to ensure trust and compliance.
- **Crisis Detection and Escalation:** A key research direction is detecting suicidal ideation or crisis language and escalating the conversation to human counselors or emergency services.

2.4 Conclusion

The use of AI in mental health support has the potential to revolutionize the accessibility and responsiveness of emotional care. Advances in NLP, emotion detection, and contextual modeling have enabled the creation of intelligent and empathetic chatbots. However, for these systems to become reliable mental health tools, ongoing research must address challenges related to data privacy, ethical considerations, fairness, and interpretability. The goal is to create safe, inclusive, and impactful digital mental health interventions that support users anytime, anywhere.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 Dataset

The dataset used for the mental health chatbot consists of three publicly available datasets: **Mental Health FAQ**, **NLP Mental Health Conversations**, and the Psychology Dataset Split. These datasets provide a variety of conversations and frequently asked questions about mental health, including topics like anxiety, depression, therapy, and emotional well-being. The diversity of the dataset ensures the chatbot can handle a wide range of user queries, although it is still limited to the scope of these pre-existing datasets.

Questions	Response
I'm feeling really anxious lately and I don't know why.	It's common to feel anxious at times, and there can be many reasons for it. Have there been any recent changes or stressors in your life that may be contributing to your anxiety? Let's work together to identify any triggers and develop coping strategies to manage your anxiety.
I'm feeling really overwhelmed with work and school. I don't know how to manage my time and it's causing me a lot of stress.	It sounds like you're going through a difficult time. Let's work on creating a schedule that prioritizes your tasks and allows for breaks throughout the day. We can also discuss some stress management techniques to help you cope with the pressure.
I'm having trouble sleeping and I'm constantly tired. I think it might be because of my medication.	It's important to talk to your doctor about any side effects you're experiencing from your medication. We can also explore some relaxation techniques to help you fall asleep and improve your sleep quality.
I've been feeling really anxious lately, and I don't know why.	It's common to feel anxious without knowing the exact cause. Would you like to explore some possible triggers or coping mechanisms that could help you manage your anxiety?

Fig.3.1 **Sample images**

3.2 Data Collection

The primary dataset used in this project is Merged2.csv, which is the merged version of all three datasets used, and it contains user questions and expert responses related to mental health. The dataset includes two main columns: Questions and Responses. This dataset was manually curated and serves as a knowledge base for similarity-based responses.

3.3 Data Preprocessing

Data preprocessing is crucial to ensure the chatbot processes and responds to user inputs effectively. To ensure accurate natural language understanding, the text data was cleaned and normalized using the following NLP preprocessing techniques:

- **Data Cleaning:** The dataset was cleaned by removing any duplicates and null values. This ensures that only valid and unique data is included for training.
- **Text Tokenization:** The text data was tokenized, breaking down sentences into individual words to convert the text into a numerical format that a machine learning model can understand.
- **Text Padding:** Padding was added to the tokenized data to ensure that all input sequences had the same length, allowing for consistent input into the model.
- **Text Vectorization:** The tokenized and padded data was vectorized to transform the text into a numerical format suitable for input into the machine learning model.
- **Punctuation Removal:** All punctuation characters were stripped using regular expressions.
- **Lowercasing:** All text was converted to lowercase to ensure case-insensitive matching.

Multithreading was applied using Python's ThreadPoolExecutor to parallelize preprocessing across all records for improved performance.

The model utilizes an external API key, which is used to generate responses for queries not covered by the dataset, enhancing its ability to provide helpful answers for queries outside the scope of its training.

```

leematizer=WordNetLemmatizer()
for index,row in data.iterrows():
    filter_sentence=[]
    # Access the original question string, assuming it's stored as a string
    sentence=str(row['Questions'])
    sentence=re.sub(r'^\w\s','',sentence)
    words=nlk.word_tokenize(sentence) #for Feature Extraction
    words=[w for w in words if not w in stop_words]
    for word in words:
        filter_sentence.append(leematizer.lemmatize(word))
    print(filter_sentence)
    # Use .loc instead of .ix for label-based indexing
    data.loc[index,'Questions'] = filter_sentence

```

Fig.3.2 Features Used

3.4 Model Architecture

3.4.1 Architecture

The chatbot follows a hybrid rule-based retrieval architecture, outlined as follows:

1. Crisis Keyword Detection
2. Greeting and Farewell Recognition
3. TF-IDF Vectorization of both user input and dataset questions
4. Cosine Similarity Matching to find the best answer
5. Fallback to Gemini API when the similarity score is low or the answer is emotionally inadequate

3.4.2 Key Components

- Preprocessing Module: Cleans, tokenizes, and lemmatizes incoming text.
- Crisis Word Detector: Scans for predefined keywords related to mental health emergencies.
- Intent Classifier: Identifies if the message is a greeting or farewell.
- TF-IDF Vectorizer: Converts the user message and dataset entries into TF-IDF vectors
- Similarity Engine: Uses cosine similarity to compare input against existing questions.
- Threshold Logic: Determines whether to return a dataset response or fallback.
- Response Handler: Returns final output, including emergency help, greeting, or fetched answer.

3.5 Code Implementation

The implementation of the chatbot is carried out using Python in Google Colab. The key components of the code include:

1. Data Preprocessing:
 - The input data (user queries) is cleaned, tokenized, and padded to prepare it for the model.
 - Text vectorization is applied to convert the tokenized text into numerical data suitable for input into the model.
2. Model Creation and API Integration:

- The chatbot processes user input through a simple classification system.
 - If no match is found in the dataset, the model queries the external API to generate a response.
3. User Interface using Gradio:
- Gradio is used to create an interactive web interface for the chatbot. It allows users to enter mental health-related queries and receive responses from the chatbot in real-time.
 - Gradio provides an easy-to-use interface that allows the user to enter text and receive the chatbot's response, either from the dataset or the external API.

3.6 Deployment

The chatbot is deployed using Gradio, which provides a user-friendly interface for interacting with the system. Key aspects of the deployment include:

- Frontend:
Gradio offers a simple and responsive interface for users to enter mental health-related queries. Users can type their questions, and the chatbot responds based on the available data or via API integration.
- Backend:
The backend processes the input data, performs classification, and either fetches a predefined response from the dataset or queries the API for an answer.
- Deployment on Google Colab:
The entire chatbot model is deployed on Google Colab, which ensures efficient computation and scalability. The model runs in the cloud, and users can interact with it seamlessly through Gradio.

3.7 Limitations

- **Contextual Understanding:** The current model does not retain context between different user inputs. This limits its ability to handle multi-turn conversations effectively.
- **Dataset Limitations:** While the chatbot performs well on queries within the dataset, its performance may degrade for very specific or rare queries. The external API helps mitigate this limitation by providing more comprehensive responses.
- **No LLM Integration:** As it does not use advanced language models, responses may lack contextual depth or flexibility.
- **No Sentiment Analysis:** The chatbot does not adapt its tone or strategy based on emotional cues from the user.
- **Limited Generalization:** It cannot handle nuanced or complex conversations outside its training scope.
- **No Real-Time Learning:** The model does not improve or update based on user interactions over time.

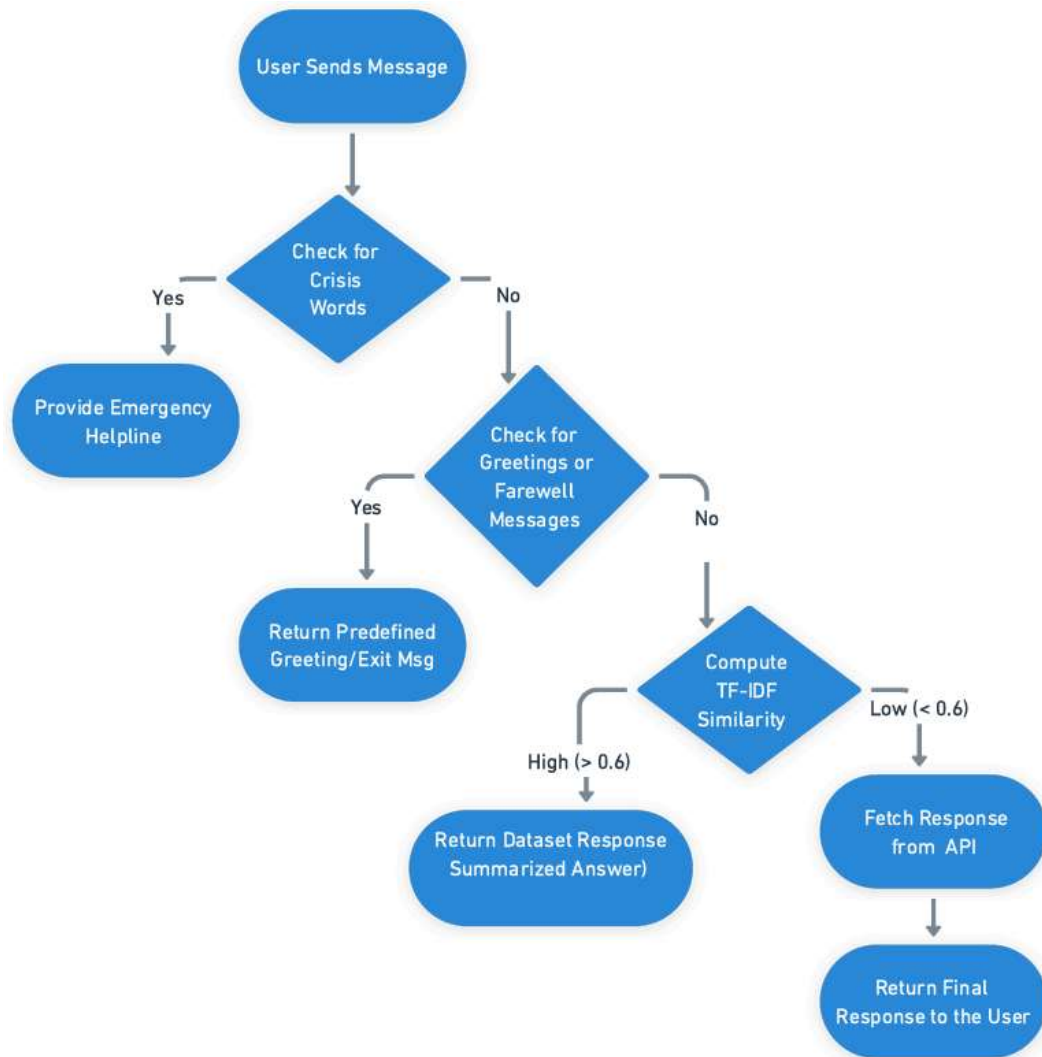


Fig 3.6 System Architecture

CHAPTER 4

IMPLEMENTATION

4.1 Technology Stack :

Technology	Function	Logos
Python	For model development and to build the backend	
Google Colaboratory	To run and execute models using Python	
Gradio	To build an interactive user interface.	
Excel	Used for reading, filtering, and managing the CSV/Excel dataset of FAQs.	

Gemini API	Used to connect with external services.	
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Table 4.1 Technology Stack

4.2 Libraries Used

The following libraries were essential for implementing the mental health chatbot project, enabling tasks ranging from data preprocessing to UI development and intelligent response generation:

- **NumPy**: Provides support for handling large, multi-dimensional arrays and matrices, although it is used minimally in this project.
- **Pandas**: Used for loading, manipulating, and processing structured data from CSV files, particularly the mental health question-response dataset.
- **Matplotlib** (*optional*): Useful for visualizing model-related metrics such as TF-IDF similarity matrices, word frequencies, or keyword detection trends.
- **Gradio**: Enables the creation of an interactive, browser-based user interface that allows users to chat with the mental health chatbot.
- **NLTK (Natural Language Toolkit)**: Used for core natural language processing tasks including tokenization, stop word removal, punctuation cleaning, and lemmatization.
- **Scikit-learn (sklearn)**:
 - `sklearn.feature_extraction.text.TfidfVectorizer`: Converts the dataset prompts and user queries into TF-IDF vectors for similarity comparison.
 - `sklearn.metrics.pairwise.cosine_similarity`: Calculates cosine similarity scores between vectorized queries and the dataset to identify the most relevant responses.
- **google.generativeai**: Provides integration with Google's Gemini API, which is used to generate fallback responses when the TF-IDF similarity is too low or the input is emotional in nature.
- **concurrent.futures.ThreadPoolExecutor**: Allows preprocessing (e.g., tokenization, stop word removal) to be applied to multiple dataset rows in

parallel, speeding up the execution.

- **re and string (Python built-in libraries):** Used for cleaning and formatting raw text input, specifically removing punctuation and special characters.

4.3 Implementation

The implementation process for the mental health chatbot project involved several stages, starting from data collection and preprocessing to model training and deployment.

4.3.1 Data Collection

The dataset used for this chatbot consists of mental health-related conversations and advice. Data collection focused on gathering queries related to common mental health topics, including anxiety, depression, stress, and general well-being.

The dataset Merged2.csv was formed by combining three datasets:

- **Mental Health FAQ** (Kaggle)
- **NLP Mental Health Conversations** (Kaggle)
- **Psychology Dataset Split** (HuggingFace)

The CSV contains:

- Questions — the original mental health queries.
- Response — full-length expert replies.
- A new column Summarized_Response — generated using sentence-level TF-IDF.

```
import pandas as pd
data = pd.read_csv("/content/Merged2.csv")
```

Fig 4.3.1 Data Import

4.3.2 Data Preprocessing:

Data preprocessing was a critical step to normalize, clean, and prepare the input data for accurate semantic matching. It involved the following stages:

- **Stop Word Removal:** Removed commonly used words such as "the", "is", and "in" using NLTK's stopwords list.

- Punctuation Removal: Used regular expressions (re) to strip punctuation marks.
- Lowercasing: All text was converted to lowercase to ensure uniform matching.
- Tokenization: Input text was split into individual tokens using `nlk.word_tokenize()`.
- Lemmatization: Used `WordNetLemmatizer` to convert words to their base form (e.g., "running" to "run").
- Multithreading: Employed `ThreadPoolExecutor` to process dataset rows in parallel and improve speed.

```
# Initialize NLP tools
stop_words = set(stopwords.words('english'))
lemmatizer = WordNetLemmatizer()

# Function to tokenize and clean text
def preprocess_text(text):
    text = re.sub(r'^\w\s]', '', str(text)) # Remove punctuation
    words = word_tokenize(text.lower()) # Tokenize and convert to lowercase
    words = [lemmatizer.lemmatize(w) for w in words if w not in stop_words] # Remove stopwords & lemmatize
    return " ".join(words) # Return cleaned text

# Apply tokenization using multi-threading
with ThreadPoolExecutor() as executor:
    data['Tokenized_Questions'] = list(executor.map(preprocess_text, data['Questions']))
    data['Tokenized_Response'] = list(executor.map(preprocess_text, data['Response']))
```

Fig 4.3.2 Data Preprocessing

4.3.3 Model Architecture

The chatbot uses a rule-based and retrieval-based hybrid architecture:

1. Crisis Detection: Checks for emergency phrases like "suicide", "want to die".
2. Greeting/Farewell Detection: Detects common phrases like "hi" or "bye".
3. TF-IDF Similarity: Compares user query to dataset questions.
4. Gemini API Fallback: Used if similarity < 0.6.

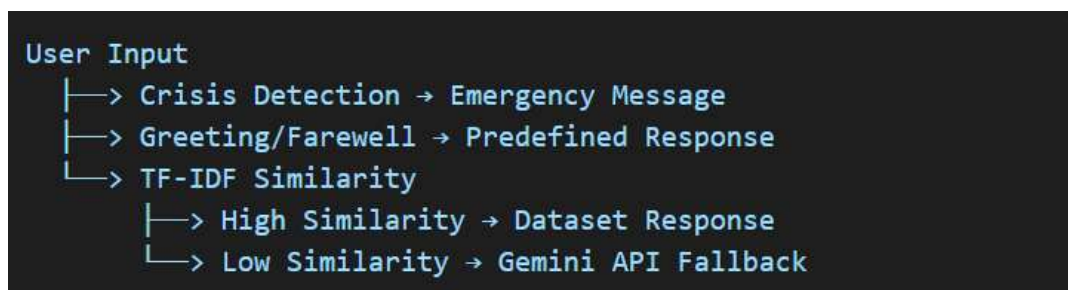


Fig 4.3.3 Pipeline Overview

4.3.4 Model Compilation and Training

This chatbot does not involve neural network training. It uses TF-IDF to convert text data into numerical vectors and cosine similarity to find the most relevant response.

```
# NLP-based response using dataset
user_tfidf = TfidfVec.transform([message])
vals = cosine_similarity(user_tfidf, tfidf_matrix)
idx = vals.argsort()[0][-1] # Get the best matching index
flat = vals.flatten()
flat.sort()
req_tfidf = flat[-1]

# Threshold-based response filtering
SIMILARITY_THRESHOLD = 0.6
```

Fig 4.3.4 Response Fetching

4.3.5 Deployment

Deployment is done using Gradio, allowing anyone to interact with the chatbot via a web browser. The interface consists of a single input textbox and an output response field.

```
# Gradio UI setup
with gr.Blocks() as app:
    gr.Markdown("# 🌱 Thrive Mental Health Chatbot 🗨️")
    gr.Markdown("A chatbot for mental well-being support. ")

    chatbot = gr.ChatInterface(fn=chatbot_response, title="Thrive Mental Health Chatbot", type="messages")

    gr.Markdown("⚠️ This chatbot provides general advice. Seek professional help for serious issues.")
app.launch(share=True, debug=True)
```

Fig 4.3.5 Model Layer 2

4.4 User Interface

The Gradio Interface includes:

- **Input Text Box:** Users can enter questions or concerns related to mental health.
- **Submit Button:** Sends input to the model for classification.
- **Output Display:** Shows the chatbot's response based on the detected intent.

The interface is user-friendly and designed to provide quick, supportive responses to user queries.

4.5 Visualizations

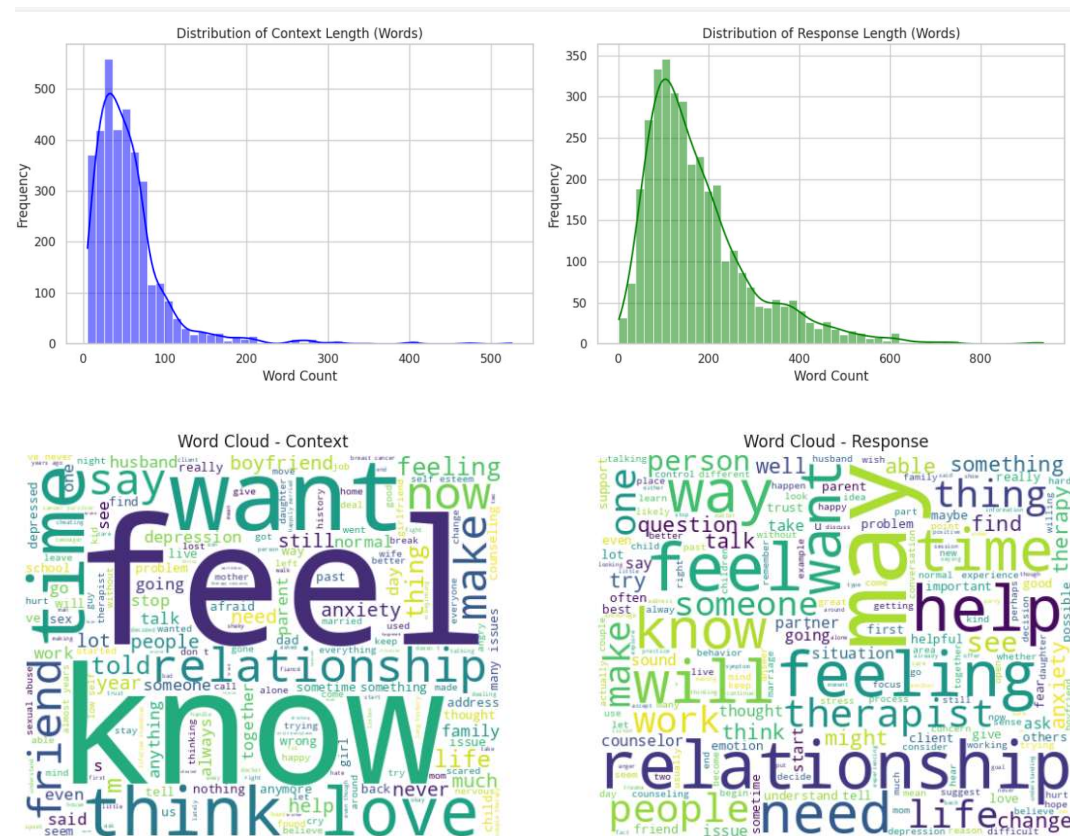


Fig 4.5 Visualizations

CHAPTER 5

RESULT AND DISCUSSION

5.1 System Output and Functionality

The mental health chatbot was developed using a rule-based NLP pipeline with a TF-IDF + cosine similarity approach, followed by fallback generation via Google Gemini API. The dataset was prepared by merging three public datasets (Mental Health FAQ, NLP Mental Health Conversations, and Psychology Dataset Split) and cleaned using Excel to remove null and duplicate entries.

Once deployed in Google Colab, the chatbot accepts user queries through a Gradio UI interface. Upon receiving input, the system performs the following steps:

- Step 1: Preprocessing – Input is cleaned (lowercased, punctuation removed, lemmatized).
- Step 2: Similarity Matching – Uses TF-IDF vectorization and cosine similarity to find the most similar question in the dataset.
- Step 3: Response Retrieval or Generation:
 - If similarity > threshold (e.g., 0.7): returns the dataset-based response.
 - Else: Calls the Gemini Pro API using an API key to generate a fresh response.

Additional functionality:

- Crisis keyword detection (e.g., "suicide", "depression") triggers a message with emergency helpline info.
- Lightweight and browser-accessible via Gradio, making the tool deployable for quick demonstrations or academic use.

Despite being a stateless system, the chatbot serves as a helpful first-level assistant for users seeking general mental health support.

5.2 Practical Implications

The chatbot functions as a hybrid mental health assistant, offering:

- Reliable FAQ-style answers from its dataset.
- Dynamic, up-to-date responses through Gemini API fallback.
- Private, non-judgmental environment for mental health inquiries.

Its ease of deployment and low resource requirements (runs entirely in Colab) make it highly applicable for:

- Mental health awareness websites,
- College counseling platforms,
- Educational and training purposes.

By helping users express concerns and get basic information, it promotes emotional openness and reduces mental health stigma.

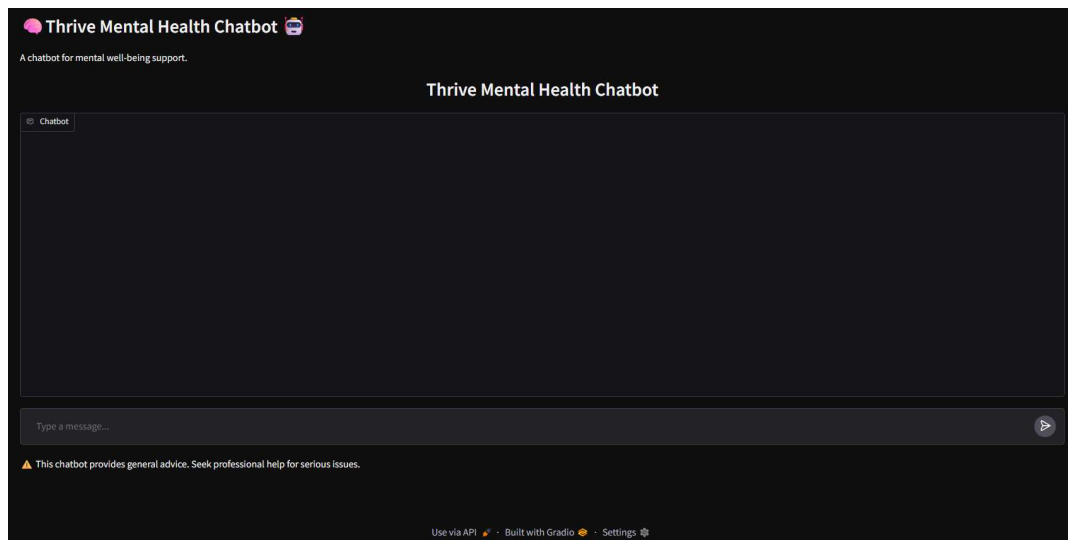


Fig 5.2.1 User Interface

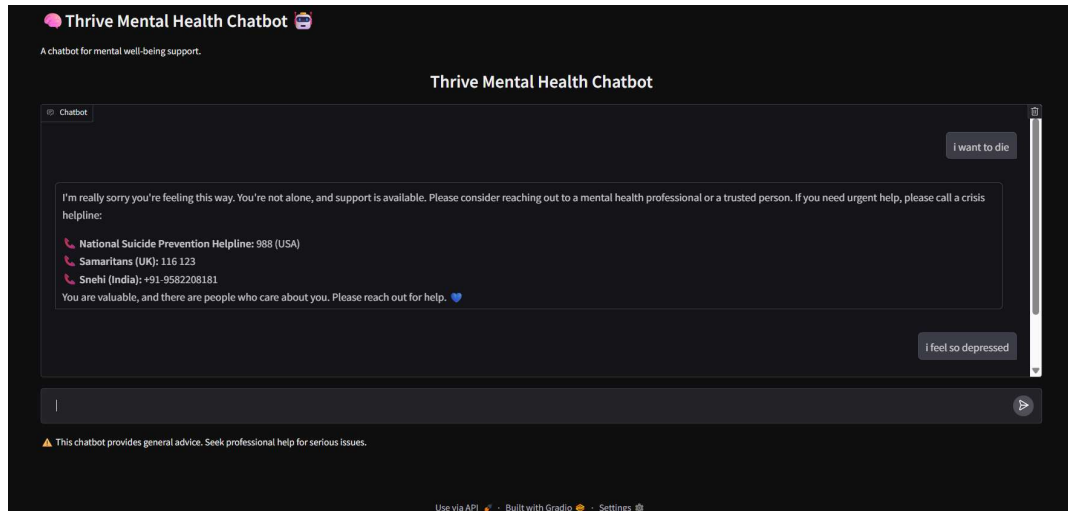


Fig 5.2.2 Questions and Response

5.3 Limitations

While functional and flexible, the current version has the following limitations:

- **No Contextual Memory:** Each question is treated independently; the system cannot carry information across multiple turns in a conversation.
- **No Quantitative Evaluation:** The system has not been tested using NLP metrics (accuracy, BLEU, F1). Performance is judged manually.
- **Limited Dataset Coverage:** If the question is too different from training data, the Gemini API handles it, but the dataset itself is narrow in scope.
- **Not Clinically Validated:** This chatbot is not approved by mental health professionals and should not be used in crisis situations.
- **Dependency on API:** For non-matching queries, a working Gemini API key and internet connectivity are required.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

This project successfully demonstrates a hybrid chatbot system combining traditional NLP methods (TF-IDF) with modern LLM-based response generation (Gemini API). Using cleaned mental health datasets, the chatbot is capable of delivering accurate and supportive responses to common queries, with a dynamic fallback for handling complex or unseen questions.

Despite lacking conversational memory or deep learning training, it remains a robust prototype for mental health support. The chatbot promotes digital wellness and creates a foundation for more advanced AI tools in emotional and psychological support.

Future Scope

The project can be improved significantly through the following enhancements:

- Add Conversation Memory: Use Transformers (e.g., DialoGPT, BERT) or LLMs with memory to support context-aware, multi-turn conversations.
- Trainable Deep Models: Fine-tune pre-trained language models on merged datasets to increase response quality.
- Feedback Loop: Let users rate responses to guide improvement using reinforcement learning or supervised retraining.
- Multilingual Support: Support for Hindi, Marathi, or regional languages to improve accessibility.
- Emergency Escalation System: Integrate local helpline APIs or chatbot escalation logic when a crisis is detected.
- Mobile App Deployment: Convert the Gradio interface into a mobile app using Flutter or React Native for real-time accessibility.
- Voice Input/Output: Add speech recognition and TTS for better inclusivity.

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